Khalil LAGHA

M2 student in Intelligent Systems for Automotive & Aerospace (SIAA) — embedded systems, control & power electronics · Université Paris-Saclay · Work-study via CFA EVE

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PROFILE

M2 student in Intelligent Systems for Automotive & Aerospace (SIAA, Université Paris-Saclay, **work-study via CFA EVE**), **ranked 2nd of the M1 EEA cohort (15.43/20)**. Specialized in **control and embedded systems**: field-oriented control (**FOC**), **PID** controllers, observers, **Kalman filter**, **Matlab/Simulink** validation (specification → implementation → test) then real-hardware testing — across **five internships** from the lab (GIPSA-lab) to industrial sites. Target fields: **EV / powertrain**, **embedded systems**, **applied AI**. **Seeking: work-study contract (alternance) from September 2026 — 2 days company / 3 days university.**

WORK EXPERIENCE

Research Intern — *GIPSA-lab, Grenoble* 04–07/2026 · 3.5 months

- Modeled a **PEMFC fuel cell** in Matlab/Simulink — **specification → implementation → test** approach, **6 %** dynamic and **1 %** static error against real measurements.
- Tuned the **regulation loop** (PI controller) of a **DC-DC boost converter** — validated in simulation then on a real test bench (tests, measurements, documentation).

Intern — *SLB / Schlumberger* 07/2025 · 1 month

- Authored technical summaries **in English** (directional drilling) in an international industrial environment; earned the **NEST & SIPP Level 2** HSE certifications.

Power Electronics & Automation Intern — *EURL Lagha* 01/2024 · 15 days + 09/2024 · 1 month

- Worked on the electronics of real industrial equipment: a **DC-DC buck converter** (50 V → 0–48 V adjustable, **25 A**) and **4 transformer-rectifiers** (100 V DC / 80 A ≈ 8 kW, ripple < 5 %).
- Contributed to the supervision HMI (**Siemens TIA Portal**, Ladder PLC); configured industrial control applications.
- Produced the electrical schematics in **EPLAN** — direct interface with manufacturing; studied a transformer against a customer specification.

Power Grid Intern — *Sonelgaz* 03/2024 · 15 days · observation

- Studied the full generation → distribution chain (**400/220 kV → 60 kV → 30/10 kV → 400/230 V**) and substation architecture; visited an MV/LV substation.

EDUCATION

M2 SIAA — Intelligent Systems for Automotive & Aerospace · *Université Paris-Saclay (Évry) · CFA EVE* 09/2026 · admitted · work-study

M1 EEA (EECS) · *Université Grenoble Alpes* 2025 – 2026

Ranked 2nd of the cohort · annual average **15.43/20**.

Engineering programme in Electrical Engineering · *École Nationale Polytechnique, Algiers* 2021 – 2025

4 of 5 years completed · moved to France for the master's · prep classes: among the top-ranked.

Baccalauréat in Mathematics 2021

Highest honours (Très Bien) — **16.93/20**.

PROJECTS

Field-oriented control (FOC) of an induction motor — Matlab/Simulink FLAGSHIP

- Implemented Park transforms, PID controllers, observers and a **Kalman filter**; tuned the **cascaded current- and speed-regulation loops** — the core of electric-vehicle traction.

PWM generator with adjustable frequency and duty cycle — 555 timer + D flip-flops

- Designed a **PWM** generator with **adjustable frequency and duty cycle**; simulated in **Proteus** then **built and fine-tuned on real hardware** — a complete electronics-diagnostics workflow.

Neural network from scratch (MLP) — Matlab · open source

- Hand-coded a multilayer perceptron and its **backpropagation** without any toolbox; debugged, tested and **published under the MIT license** — github.com/KhalilEllahLAGHA/mlp-backpropagation-matlab.

Electrical machine sizing — Python · team

- Built a **Python (OOP, PyQt)** application — from specification to automatic parameter generation; finite-element validation (**FEMM**).

Also: cascade control of a DC motor — three-phase thyristor bridge (Simulink); Siemens Step7 automation (HMI + Ladder); Arduino line-follower robot (Poly Maze competition); power grid analysis (load flow, 3–118 buses, Matlab).

TECHNICAL SKILLS

Control engineering: PID, LQR, observers, Kalman filter · field-oriented control (FOC) · cascade control · regulation-loop tuning · system identification

Embedded systems & programming: C/C++, Python (OOP, PyQt), Arduino (IDE + firmware) · hands-on electronics, from schematic to real circuit (Proteus, LTspice)

Industrial automation: Siemens PLC (TIA Portal, Step7 — Ladder, SCL, FBD, SFC) · HMI/SCADA · Grafcet

AI: neural networks (backpropagation), genetic algorithms, data fusion, fuzzy logic

Power electronics: DC-DC conversion (buck, boost), AC-DC, DC-AC, AC-AC · PWM · power electronics for machine control

SOFTWARE

Matlab/Simulink · Python (OOP, PyQt) · C/C++ · Arduino · Proteus · LTspice · Siemens TIA Portal · Step7 (Ladder/SCL) · EPLAN · AutoCAD Electrical · FEMM · SolidWorks · Git/GitHub

SOFT SKILLS & ACTIVITIES

Fast learner · comfortable with new tools & software · problem solving · rigorous · organized — IEEE Student Branch ENP · VIC (ENP)

LANGUAGES

French — C1 · English — C2 · Arabic — native

CERTIFICATIONS

SLB — NEST & SIPP Level 2 (07/2025) — HSE, industrial operations safety